

Greatest Number

Tags:

Time Limit: 2 s Memory Limit: 64 MB
Submission: 197 AC: 67 Score: 98.57

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AI Code Tips

Description

Saya likes math, because she think math can make her cleverer.

One day, Kudo invited a very simple game:

Given N integers, then the players choose no more than four integers from them (can be repeated) and add them together. Finally, the one whose sum is the largest wins the game. It seems very simple, but there is one more condition: the sum shouldn't larger than a number M .

Saya is very interest in this game. She says that since the number of integers is finite, we can enumerate all the selecting and find the largest sum. Saya calls the largest sum Greatest Number (GN). After reflecting for a while, Saya declares that she found the GN and shows her answer. Kudo wants to know whether Saya's answer is the best, so she comes to you for help.

Can you help her to compute the GN?

Input

The input consists of several test cases.

The first line of input in each test case contains two integers N ($0 < N \leq 1000$) and M ($0 < M \leq 1000000000$), which represent the number of integers and the upper bound.

Each of the next N lines contains the integers. (Not larger than 1000000000)

The last case is followed by a line containing two zeros.

Output

For each case, print the case number (1, 2 ...) and the GN.

Your output format should imitate the sample output. Print a blank line after each test case.

Samples

input

Copy

2 10

```
100
2
```

```
0 0
```

output [Copy](#)

Case 1: 8

Source

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